

Kareem Abdelmaqsoud

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INTERESTS

AI for Science, Material Science, Catalysis, Machine Learning, Large Language Models (LLMs), and Data Science

EDUCATION

Carnegie Mellon University

PhD in Chemical Engineering – GPA 3.96

Pittsburgh, PA
August 2022 – December 2026

Research Advisors: John R. Kitchin & Andrew J. Gellman

Selected Coursework: Data science & ML – Deep learning – AI Agents

Industry collaborations: Meta Fundamental AI Research (Fairchem) team

University of Rochester

Bachelor of Science - Chemical Engineering – GPA 3.95

Rochester, NY
August 2018 - May 2022

Research Advisor: Andrew D. White

Selected Coursework: ML for Molecules and Materials

CURRENT RESEARCH PROJECTS

Agentic Workflow for Improving Electrochemical Interfaces

Carnegie Mellon University

February 2026 – current

- Built an LLM agentic workflow for autonomous surface-doping optimization of water–solid electrochemical interfaces using molecular dynamics (MD).
- Integrated LangGraph, literature search, LAMMPS simulations, and ChromaDB database, significantly improving hydrogen bonding in 4 of 9 tested systems.

Computational Design of Rare-earth Metal Alloys

Carnegie Mellon University

August 2024 – current

- Built an ML-guided materials discovery pipeline reducing experimental search space by 10× and enabling discovery of a ductile tungsten alloy validated experimentally.
 - Developed interpretable physics-based model linking electronic structure features to alloy ductility, enabling predictive screening of new alloys.
 - Built automated high-throughput materials simulation pipelines integrating DFT, ML potentials, and CALPHAD for phase diagram prediction.
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INDUSTRY EXPERIENCE

Bosch USA

Watertown, MA

Computational Materials Science Intern

May – August 2026

- Applying universal ML potentials to build a phase diagram to model Aluminum Germanium (AlGe) alloy for semiconductor applications in MEMS devices
- Study the effect of Si contamination on phase stability and eutectic composition of the AlGe alloy.

Orbital Materials

Princeton, NJ

Machine Learning Researcher Intern

May – August 2025

- Within a 3-month internship, I developed OrbMol, a large-scale ML potential trained on 100M organic molecular structures incorporating magnetism and charge effects.
- Achieved state-of-the-art accuracy compared with existing molecular ML potentials.
- Contributed to model architecture, large-scale dataset preparation, and training pipelines.

Entos (Now Iambic Therapeutics)

Remote

Machine Learning Engineer Intern

January – July 2022

- Performed hyperparameter tuning for the model used by the experimental drug discovery team to improve its accuracy
- Implemented new pretraining methods such as contrastive learning that led to 10-20% improvement in accuracy

SKILLS

- **Programming:** Python
 - **Frameworks:** Kubernetes, AWS, Git, MongoDB, atomate, FastMCP, and LangChain
 - **Deep Learning & Data science:** PyTorch, TensorFlow, NumPy, Pandas, and scikit-learn
 - **Packages:** ASE, Pymatgen, VASP for DFT, and CALPHAD, LAMMPS for molecular dynamics
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HONORS & AWARDS

- Richard King Mellon Foundation Presidential Fellowship in Energy.
 - 1st Place Award in the Undergraduate Poster Competition at the AIChE conference, Boston, MA.
 - Research & Innovation Grant Recipient for my undergraduate research.
 - Tau Beta Pi Engineering Honor Society.
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PUBLICATIONS

Investigating the Error Imbalance of Large-Scale Machine Learning Potentials in Catalysis (Catalysis Science & Technology)

Abdelmaqsoud, K., Shuaibi, M., Kolluru, A., Cheula, R., & Kitchin, J. R.

Structure-Sensitive Reaction Kinetics of Chiral Molecules on Intrinsically Chiral Surfaces (Journal of Phys Chem)

Abdelmaqsoud, K., Radetic, M., Fernandez-Caban, C., Widom, M., Kitchin, J. R., & Gellman, A. J.

Uncertainty Quantification in Graph Neural Networks with Shallow Ensembles (MLST)

Vinchurkar, T., Abdelmaqsoud, K., & Kitchin, J. R.

Computational Design of Ductile Additively Manufactured Tungsten-Based Refractory Alloys (Comp Mat Science)

Abdelmaqsoud, K., Sinclair, D., Karra, V. S. S. A., Taheri-Mousavi, S. M., Widom, M., Webler, B. A., & Kitchin, J. R.

Electronic Structure and Elasticity of the Ta–W Solid Solution (Physical Rev Mat)

Abdelmaqsoud, K., Kitchin, J. R., & Widom, M.

UMA: A Family of Universal Models for Atoms (NeurIPS)

Wood, B. M., Dzamba, M., Fu, X., Gao, M., Shuaibi, M., Barroso-Luque, L., Abdelmaqsoud, K., Gharakhanyan, V., Kitchin, J. R., Levine, D. S., Michel, K., Sriram, A., Cohen, T., Das, A., Rizvi, A., Sahoo, S. J., Ulissi, Z. W., & Zitnick, C. L.

The Open DAC 2025 Dataset for Sorbent Discovery in Direct Air Capture (Nature Chem Eng)

Sriram, A., Brabson, L. M., Yu, X., Choi, S., Abdelmaqsoud, K., Moubarak, E., de Haan, P., Löwe, S., Brehmer, J., Kitchin, J. R., Welling, M., Zitnick, C. L., Ulissi, Z. W., Medford, A. J., & Sholl, D. S.